

TASK ON RELATIONAL DATABASE SERVICE(RDS)

1. Create MariaDB DB on EC2.

Objective

To launch an EC2 instance and install MariaDB database on the EC2 server.

Steps Performed

Launched an EC2 instance using Amazon Linux.

Connected to the EC2 instance using EC2 Instance Connect.

Updated the installed packages using the command:

```
sudo dnf update -y
```

Installed MariaDB using:

```
sudo dnf install mariadb105-server -y
```

Started the MariaDB service:

```
sudo systemctl start mariadb
```

Enabled MariaDB to start automatically after system reboot:

```
sudo systemctl enable mariadb
```

Verified the MariaDB service status using:

```
sudo systemctl status mariadb
```

The MariaDB service was successfully running with the status:

active (running)

Result

MariaDB was successfully installed and configured on the EC2 instance.

Evidence

Screenshot:

Attach the screenshot showing the systemctl status mariadb output with active (running) status.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-0d3571e2fab9d29bd

aws [Search] [Alt+S] United States (N. Virginia) CHVK (634176595009) CHVK

EC2 > Instances > i-0d3571e2fab9d29bd

Instance summary for i-0d3571e2fab9d29bd (MariaDB-ec2) [Info](#)

[Connect](#) [Instance state](#) [Actions](#)

Updated less than a minute ago

Instance ID i-0d3571e2fab9d29bd	Public IPv4 address 54.242.220.192 open address	Private IPv4 addresses 172.31.87.133
IPv6 address -	Instance state Running	Public DNS ec2-54-242-220-192.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-87-133.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-87-133.ec2.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t2.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations
Auto-assigned IP address 54.242.220.192 [Public IP]	VPC ID vpc-0b174515695de2fd5	

CloudShell Agent Toolkit for AWS Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

```

Installed:
 mariadb-connector-c-3.10.10-1.amzn2023.0.2.x86_64
 mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64
 mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
 mariadb105-errmsg-3:10.5.29-1.amzn2023.0.1.x86_64
 mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64
 mysql-securelinux-1.0.14-2.amzn2023.0.1.noarch
 perl-DBI-1.652-1.amzn2023.0.1.x86_64
 perl-Math-BigInt-1:1.9998.39-2.amzn2023.0.2.noarch
 perl-Math-Complex-1.59-477.amzn2023.0.9.noarch
 perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64

Complete!
[root@ip-172-31-87-133 ec2-user]# sudo systemctl start mariadb
[root@ip-172-31-87-133 ec2-user]# sudo systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@ip-172-31-87-133 ec2-user]# sudo systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-09-19 08:15:10 UTC; 29s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Main PID: 27655 (mariabdd)
   Status: "Taking your SQL requests now..."
     Tasks: 11 (limit: 1113)
    Memory: 66.1M
       CPU: 437ms
    CGroup: /system.slice/mariadb.service
            └─27655 /usr/libexec/mariabdd --basedir=/usr

Sep 19 08:15:09 ip-172-31-87-133.ec2.internal mariadb-prepare-db-dir[27612]: The second is mysql@localhost, it has no password either, but
Sep 19 08:15:09 ip-172-31-87-133.ec2.internal mariadb-prepare-db-dir[27612]: you need to be the system 'mysql' user to connect.
Sep 19 08:15:09 ip-172-31-87-133.ec2.internal mariadb-prepare-db-dir[27612]: After connecting you can set the password, if you would need to be
Sep 19 08:15:09 ip-172-31-87-133.ec2.internal mariadb-prepare-db-dir[27612]: able to connect as any of these users with a password and without sudo
Sep 19 08:15:09 ip-172-31-87-133.ec2.internal mariadb-prepare-db-dir[27612]: See the MariaDB Knowledgebase at https://mariadb.com/kb
Sep 19 08:15:09 ip-172-31-87-133.ec2.internal mariadb-prepare-db-dir[27612]: Please report any problems at https://mariadb.org/jira
Sep 19 08:15:09 ip-172-31-87-133.ec2.internal mariadb-prepare-db-dir[27612]: The latest information about MariaDB is available at https://mariadb.org/.
Sep 19 08:15:09 ip-172-31-87-133.ec2.internal mariadb-prepare-db-dir[27612]: Consider joining MariaDB's strong and vibrant community:
Sep 19 08:15:09 ip-172-31-87-133.ec2.internal mariadb-prepare-db-dir[27612]: https://mariadb.org/get-involved/
Sep 19 08:15:10 ip-172-31-87-133.ec2.internal systemd[1]: Started mariadb.service - MariaDB 10.5 database server.
[root@ip-172-31-87-133 ec2-user]#

```

2. Insert some dummy data.

Objective

To create an employee table in the MariaDB database and insert dummy data.

Steps Performed

Connected to the MariaDB database using:

```
sudo mysql
```

Selected the companydb database:

```
USE companydb;
```

Created an employees table with the following columns:

```
CREATE TABLE employees (
```

```
id INT PRIMARY KEY,
```

```
name VARCHAR(50),
```

```
department VARCHAR(50),
```

salary INT

);

Inserted dummy employee records:

INSERT INTO employees VALUES

(1,'Karthik','DevOps',30000),

(2,'Rahul','AWS',35000),

(3,'Suresh','Testing',28000),

(4,'Anil','Development',40000);

Verified the inserted records using:

SELECT * FROM employees;

Result

The employees table was successfully created in the companydb database, and dummy employee data was inserted and verified successfully.

Evidence

Screenshot:

Attach the screenshot showing the SELECT * FROM employees; command and its output.

```
MariaDB [companydb]> SELECT * FROM employees;
+----+-----+-----+-----+
| id | name  | department | salary |
+----+-----+-----+-----+
| 1  | Karthik | DevOps    | 30000  |
| 2  | Rahul  | AWS       | 35000  |
| 3  | Suresh | Testing   | 28000  |
| 4  | Anil   | Development | 40000  |
+----+-----+-----+-----+
4 rows in set (0.000 sec)

MariaDB [companydb]> |
```

3. Take the backup of dummy data on EC2.

The companydb MariaDB database was backed up on the EC2 instance using the mysqldump command.

Command Used:

sudo mysqldump -u root companydb > comapanydb_backup.sql

The backup file comapanydb_backup.sql was successfully created on the EC2 instance.

Result:

The companydb database backup was successfully completed and stored as comapanydb_backup.sql on the EC2 instance.

Screenshot:

Attach the screenshot showing the backup command and the created backup file.

```
[root@ip-172-31-87-133 ec2-user]# mysql -h database-1.c23mcwocefr5.us-east-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-verify-server-cert --ssl-ca=../global-bundle.pem com
panydb < comapanydb_backup.sql
Enter password:

[root@ip-172-31-87-133 ec2-user]# ls -lh comapanydb_backup.sql
-rw-r--r--. 1 root root 2.1K Sep 19 08:27 comapanydb_backup.sql
[root@ip-172-31-87-133 ec2-user]#
```

4. Launch MariaDB RDS instance.

Amazon RDS was used to launch a managed MariaDB database instance.

Configuration:

DB Instance Identifier: database-1

Database Engine: MariaDB

DB Instance Class: db.t4g.micro

Availability Zone: us-east-1b

Port: 3306

Public Access: No

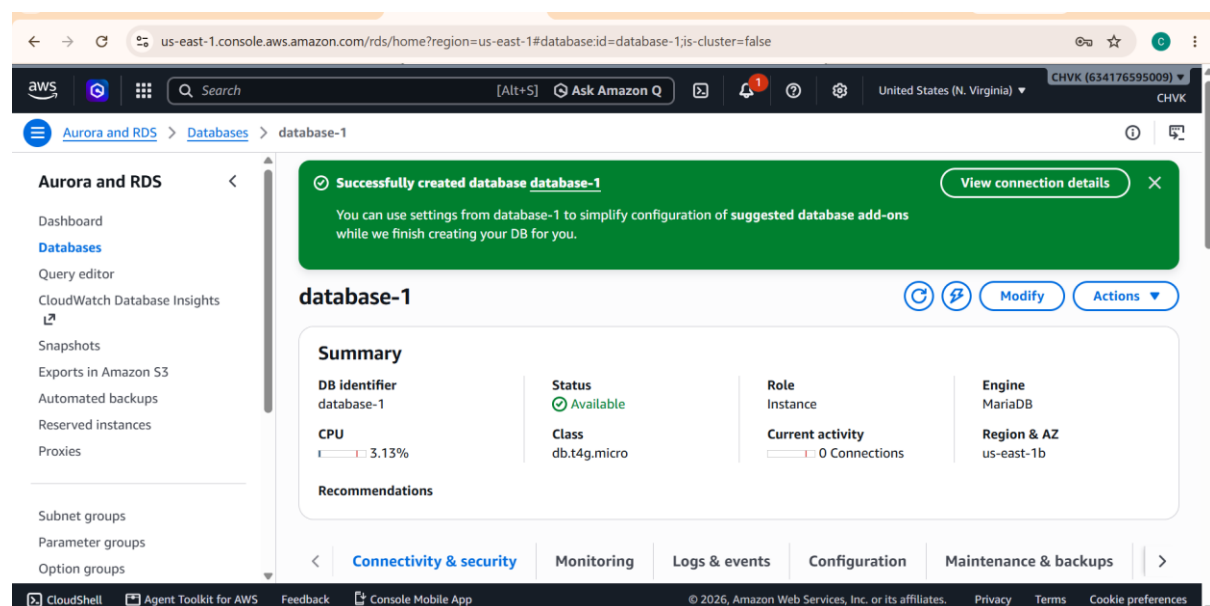
The MariaDB RDS instance was successfully created and reached the Available status.

Result:

The MariaDB RDS instance was successfully launched and configured for database migration from the EC2 instance.

Screenshot:

Attach the screenshot of the RDS database details page showing the database-1 instance with MariaDB engine and Available status.



5. Migrate database from EC2 to RDS.

The companydb database backup created on the EC2 instance was migrated to the MariaDB RDS instance.

The companydb database was created on the RDS instance and the backup file was imported successfully.

The migrated data was verified using:

USE companydb; SELECT * FROM employees;

The employees table and all 4 dummy records were successfully restored and verified on the MariaDB RDS instance.

Result:

The companydb database was successfully migrated from EC2 MariaDB to Amazon RDS MariaDB.

Screenshot:

Attach the screenshot showing the RDS companydb database and the 4 employee records.

```
[root@ip-172-31-87-133 ec2-user]# mysql -h database-1.c23mcwoefr5.us-east-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-verify-server-cert --ssl-ca=./global-bundle.pem
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 56
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE companydb
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [companydb]> SELECT * FROM employees;
+----+-----+-----+-----+
| id | name  | department | salary |
+----+-----+-----+-----+
| 1  | Karthik | DevOps      | 30000  |
| 2  | Rahul  | AWS         | 35000  |
| 3  | Suresh | Testing     | 28000  |
| 4  | Anil   | Development | 40000  |
+----+-----+-----+-----+
4 rows in set (0.001 sec)
```

6. Install MySQL DB on EC2.

MySQL Community Server was installed and configured on the EC2 instance running Amazon Linux 2023.

MySQL Version:

MySQL 8.4.11

The MySQL server was initialized and the mysqld service was started successfully.

The service status was verified and the server was running with the status active (running).

Result:

MySQL Database Server was successfully installed and is running on the EC2 instance.

Screenshot:

Attach the screenshot showing the MySQL version and mysqld service with active (running) status.

```
[root@ip-172-31-87-133 ec2-user]# mysql --version
mysql Ver 8.4.11 for Linux on x86_64 (MySQL Community Server - GPL)
[root@ip-172-31-87-133 ec2-user]#

[root@ip-172-31-87-133 ec2-user]# systemctl status mysqld --no-pager
● mysqld.service - MySQL Server
   Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-09-19 09:13:02 UTC; 20s ago
     Docs: man:mysqld(8)
           http://dev.mysql.com/doc/refman/en/using-systemd.html
   Process: 30945 ExecStartPre=/usr/bin/mysqld_pre_systemd (code=exited, status=0/SUCCESS)
   Main PID: 30972 (mysqld)
    Status: "Server is operational"
     Tasks: 35 (limit: 1113)
   Memory: 436.9M
      CPU: 1.663s
   CGroup: /system.slice/mysqld.service
           └─30972 /usr/sbin/mysqld

Sep 19 09:12:59 ip-172-31-87-133.ec2.internal systemd[1]: Starting mysqld.service - MySQL Server...
Sep 19 09:13:02 ip-172-31-87-133.ec2.internal systemd[1]: Started mysqld.service - MySQL Server.
[root@ip-172-31-87-133 ec2-user]# client_loop: send disconnect: connection reset by peer
```

7. Launch MySQL RDS image.

Amazon RDS was used to launch a managed MySQL database instance.

Configuration:

DB Identifier: database-2

Engine: MySQL Community
Instance Class: db.t4g.micro
Region: us-east-1
Availability Zone: us-east-1b
Deployment: Single-AZ
Status: Available

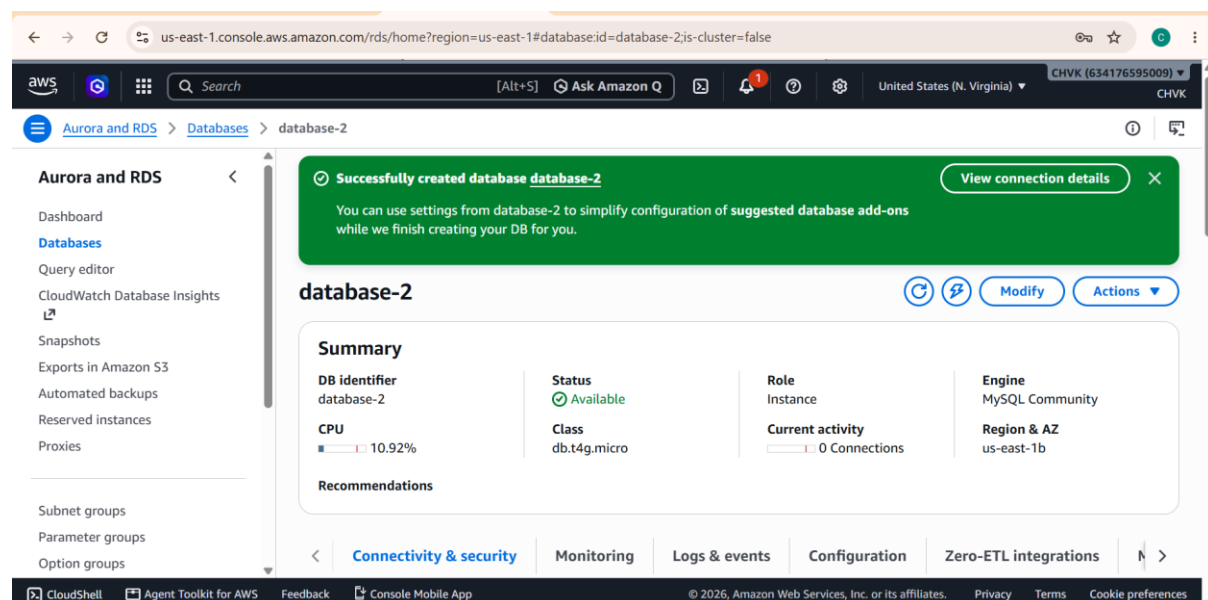
The MySQL RDS instance was successfully created and is ready for database operations.

Result:

MySQL RDS instance database-2 was successfully launched and is in Available status.

Screenshot:

RDS database-2 details page showing MySQL Community, db.t4g.micro, us-east-1b, and Available status.



8. Configure Multi-AZ.

The MySQL RDS instance database-2 was selected for Multi-AZ configuration.

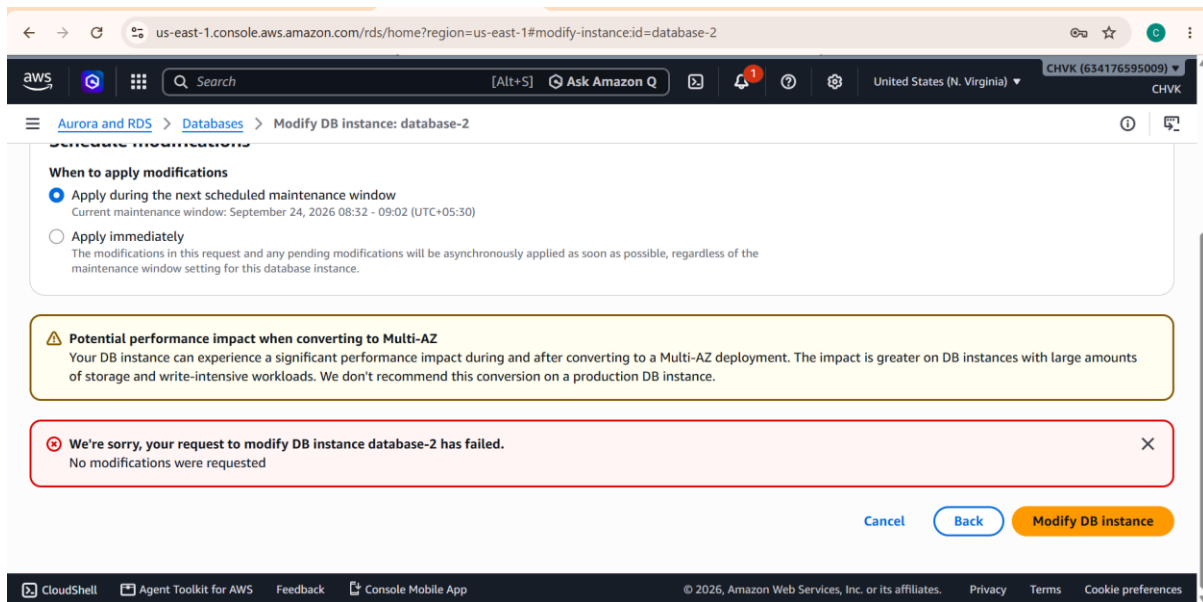
The Create a standby instance option was selected under Availability and durability. However, while applying the modification, Amazon RDS displayed the message “No modifications were requested”, and the Multi-AZ configuration could not be applied. The database instance remained in its existing configuration with Multi-AZ: No.

Result:

Multi-AZ configuration could not be completed due to the RDS modification issue.

Screenshot:

RDS Modify DB instance page showing the modification failure message “No modifications were requested.”



9. Take backup of DB and restore the DB.

A backup of the MySQL RDS database database-2 was created using an Amazon RDS DB snapshot.

Backup Configuration:

DB Instance: database-2

Snapshot Name: database-2-backup

Snapshot Status: Available

The created snapshot was successfully restored to a new MySQL RDS instance.

Restore Configuration:

Restored DB Identifier: database-2-restored

Engine: MySQL Community

Instance Class: db.t4g.micro

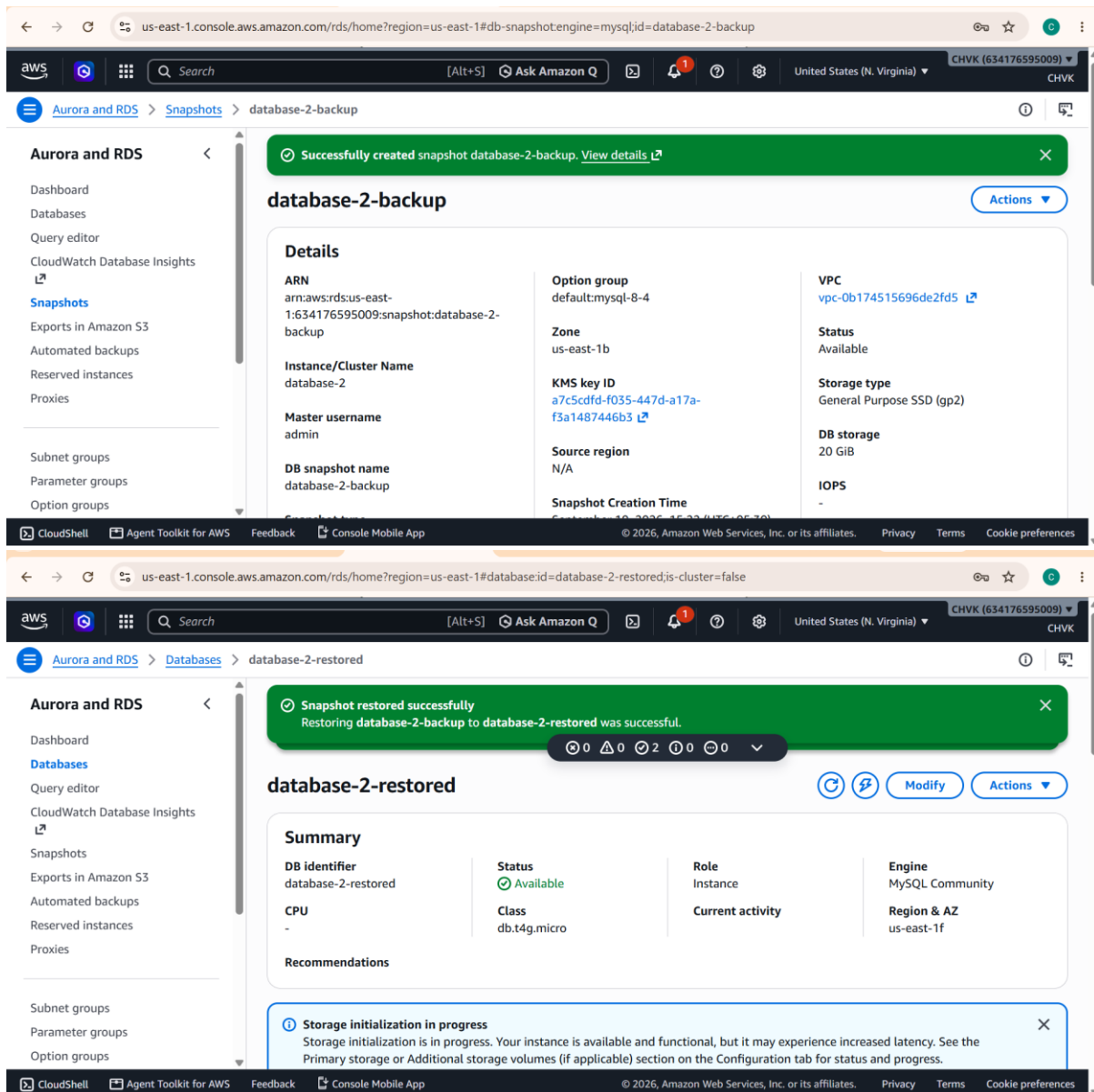
Status: Available

Result:

The MySQL RDS database was successfully backed up using a DB snapshot and restored successfully to a new RDS instance.

Screenshot:

RDS database-2-restored showing "Snapshot restored successfully" and Available status.



10. Create read replica.

A read replica was created for the MySQL RDS primary database database-2 to provide a separate database instance for read operations.

Configuration:

Primary DB: database-2

Read Replica: database-2-read-replica

Engine: MySQL Community

Instance Class: db.t4g.micro

Region: us-east-1

Availability Zone: us-east-1b

Primary Status: Available

Replica Status: Available

The read replica was successfully created and is associated with the primary database.

Result:

MySQL RDS read replica database-2-read-replica was successfully created and is available for read operations.

Screenshot:

RDS Databases page showing database-2 as Primary and database-2-read-replica as Replica, both in Available status.

The screenshot displays the AWS RDS console interface. At the top, a green banner indicates 'Successfully created replica database-2-read-replica in US East (N. Virginia)'. Below this, the 'database-2-read-replica' instance is shown with a status of 'Available'. The instance details table lists the following information:

DB identifier	Status	Role	Engine	Region	Size
database-2	Available	Primary	MySQL Co...	us-east-1b	db.t4g.m
database-2-read-replica	Available	Replica	MySQL Co...	us-east-1b	db.t4g.m

Below the table, a message states 'Storage initialization in progress' and provides a link to the 'Read Replica Status' page. The left sidebar shows the 'Aurora and RDS' navigation menu, and the bottom of the console displays the footer with copyright information and links to 'Privacy', 'Terms', and 'Cookie preferences'.